



Shri Vile Parle Kelavani Mandal's

DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING

(Autonomous College Affiliated to the University of Mumbai)

NAAC Accredited with "A" Grade (CGPA : 3.18)



Department of Computer Science and Engineering (Data Science)

Subject: Machine Learning – I (DJS23DPC402L)

AY: 2025-26

Experiment 10

(Mini Project)

Aim: Design a classifier to solve a specific problem in the given domain.

Tasks to be completed by the students:

Select a specific problem from any of the given domain areas, such as: Banking, Education, Insurance, Government, Media, Entertainment, Retail, Supply chain, Transportation, Logistics, Energy and Utility.

Task 1: Select appropriate dataset, describe the problem and justify the suitability of your dataset.

Task 2: Perform exploratory data analysis and pre-processing (if required).

Task 3: Apply appropriate machine learning algorithm to build a classifier. Perform appropriate testing of your model.

Task 4: Submit a report in the given format.

- Introduction
- Data Description
- Data Analysis
- Reason to select machine learning model
- Algorithm
- Result Analysis
- Conclusion and Future Scope.
- Python notebook



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Report on Mini Project

Machine Learning -I (DJS23DPC402L)

AY: 2025-26

TITLE OF THE PROJECT

PREDICTING USER CHURN FROM EVENT LOGS

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Guided By

Prof. Shraddha More



CHAPTER 1: INTRODUCTION

In the rapidly evolving landscape of digital platforms and subscription-based services, user churn the phenomenon of customers discontinuing their use of a service represents one of the most critical challenges for product and data teams alike. Retaining an existing customer is significantly more cost-effective than acquiring a new one, and even a marginal reduction in churn rate can translate into substantial revenue preservation. As a result, the ability to predict which users are likely to disengage has become a high-priority objective for modern data-driven organizations.

Traditional churn prediction models rely primarily on aggregated tabular features such as login frequency, subscription tier, or purchase history. While these approaches provide reasonable predictive power, they fail to capture the rich, sequential, and relational dimensions of user behavior embedded in raw event logs. Event logs record every interaction a user performs within a platform like page visits, feature clicks, session initiations, collaborative actions and contain a wealth of temporal and structural signals that aggregate metrics often obscure or discard entirely.

This project addresses the binary classification problem of predicting user churn from event log data. Given a dataset of timestamped user events the goal is to build a classifier that for each active user, predicts whether they will churn within a defined observation window. A user is labeled as churned if they cease all platform activity beyond a threshold period following their last observed event. The techniques developed herein are broadly applicable to any platform with event-level tracking: online education systems, banking portals, media streaming platforms, and enterprise software suites.



CHAPTER 2: DATA DESCRIPTION

The dataset used in this project, `user_retention.parquet`, is a parquet-format file of approximately 50 MB obtained from a real-world SaaS platform's event tracking system. It contains granular, timestamped records of user interactions, where each row represents a single event performed by a user at a specific point in time. This event-level structure preserves the detailed sequence of user actions, making it highly suitable for analyzing behavioral patterns and predicting churn.

The dataset contains over 20 columns spanning user identifiers, event metadata, geographic information, device information, and session context. The primary columns relevant to this study include:

Column	Type	Description
<code>distinct_id</code>	String	Unique identifier per user session/device
<code>person_id</code>	String	Canonical user identity across sessions
<code>event</code>	String/Categorical	Name/type of the user action (e.g., page view, click, feature use)
<code>timestamp</code>	Datetime (UTC)	Exact UTC timestamp of the event occurrence
<code>created_at</code>	Datetime	Record creation timestamp in the data warehouse
<code>prop_\$pathname</code>	String	URL path or page location where event occurred
<code>prop_\$browser</code>	String	Browser type used by the user
<code>prop_\$os</code>	String	Operating system of the user's device
<code>prop_\$geoip_count ry_code</code>	String	Country inferred from IP address via GeoIP lookup



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The dataset includes a wide range of features covering user identity, event metadata, device information, and geographic attributes. Key columns include `distinct_id` and `person_id` for uniquely identifying users, `event` for capturing the type of user action, and `timestamp` for recording when the interaction occurred. Additional contextual information such as browser type, operating system, and country code provides further insights into user behavior across different environments.

In terms of data quality, the dataset spans multiple months and includes thousands of unique users and a diverse set of event types. While most core columns are complete, some property fields such as browser, OS, and geo-location contain missing values, which is typical in event tracking systems due to factors like privacy restrictions or incomplete logging. However, no fully duplicate or empty records were found after preprocessing, ensuring reliability for downstream analysis.

Overall, the dataset is well-suited for churn prediction tasks because it captures fine-grained temporal behavior rather than aggregated summaries. Its structure allows for advanced feature engineering, including session reconstruction and behavioral profiling. Additionally, the presence of user identifiers and event relationships enables the construction of interaction graphs, making it possible to incorporate graph-based learning techniques alongside traditional machine learning models.



CHAPTER 3: DATA ANALYSIS

The data analysis phase began with a structured exploratory data analysis (EDA) to understand the dataset's dimensions, schema, and overall structure. The dataset consisted primarily of event-level records with object columns for event types and properties, and datetime columns for timestamps. Since raw event logs contain very few numerical features, this step highlighted the need for extensive feature engineering. Sample data inspection and data type verification ensured consistency, while initial checks confirmed that the dataset contained thousands of users and a wide variety of interaction events.

The next step focused on data quality assessment and missing value analysis. It was observed that certain property columns such as browser type, operating system, and geographic location contained missing values, which is typical in client-side event tracking systems due to privacy settings or incomplete logging. However, no fully empty rows or duplicate records were present after cleaning. This ensured that the dataset maintained high integrity and could be reliably used for downstream processing without introducing significant bias.

A detailed temporal analysis was then performed using the timestamp fields to understand the time span and activity distribution of users. The dataset covered a multi-month observation window, allowing meaningful tracking of user engagement over time. Key temporal features such as first activity date, last activity date, and total active duration were derived. These features were essential in capturing user lifecycle behavior and provided the basis for defining churn through recency analysis.

The analysis then progressed to session pattern reconstruction, which is a critical step when working with event-level data. Using a 30-minute inactivity threshold, user events were grouped into sessions. This enabled the computation of rich session-level features such as total sessions per user, average and median session duration, events per session, event density, and inter



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session gaps. Additionally, “deep work sessions” were identified as sessions with high duration and activity density, and a deep work ratio was computed to quantify high-quality engagement.

Following session analysis, primary success metrics and user-level aggregates were derived. These included total event count, number of unique event types, and days active on the platform. Building on these, advanced behavioral fingerprinting was performed to generate higher-level features such as power user score, struggle score (indicating repeated failures), sequence diversity (entropy of event usage), collaboration ratio, and team-oriented score. These features provided a multi-dimensional representation of user behavior beyond simple activity counts.

To incorporate relational insights, a user interaction graph was constructed as described in the PDF. In this graph, each node represents a user and edges represent shared activity patterns or co-occurrence in similar contexts, resulting in thousands of nodes and tens of thousands of edges. This enabled the extraction of graph-based features such as node degree, influence score, and neighbor churn rate. These features capture the social and network effects influencing user engagement, which are not observable in traditional tabular analysis.

Finally, churn labeling was performed using survival analysis principles. An analysis date was defined based on the latest activity in the dataset, and users were classified as churned if they remained inactive beyond a predefined threshold period. The overall churn rate observed was approximately 40–50%, reflecting realistic SaaS churn behavior. This final step converted the dataset into a well-structured format suitable for modelling, enriched with temporal, behavioral, and relational features

To capture social and relational dynamics, a user co-activity graph was constructed. Nodes in the graph represent unique users (~5,400 nodes). Edges represent shared activity contexts users who performed similar event sequences within overlapping time windows were connected, yielding approximately 40,000 edges. Each node was initialized with behavioral feature vectors derived from the fingerprinting step. This graph served as the input to the GraphSAGE model.



CHAPTER 4: DATA MODELLING

The data modelling phase focuses on developing a robust predictive framework to identify users likely to churn by leveraging both behavioral and relational signals. Unlike conventional models that treat users independently, this project adopts a hybrid modelling approach that integrates Graph Neural Networks (GNNs) with ensemble machine learning techniques. This approach is motivated by the understanding that user engagement is influenced not only by individual activity but also by interactions within a network of users.

The first stage of the modelling pipeline involves transforming raw event logs into structured user-level features through extensive feature engineering. Session reconstruction techniques are applied to group user activities into meaningful sessions, from which metrics such as total sessions, average session duration, and event density are derived. Additionally, higher-level behavioral indicators such as power user score, struggle score, sequence diversity, collaboration ratio, and team-oriented score are computed to capture nuanced engagement patterns.

To incorporate relational information, a user interaction graph is constructed where each node represents a user and edges represent shared activity patterns or co-occurrence in similar contexts. This graph captures the social and behavioral dependencies among users. A GraphSAGE model is then trained on this graph using a link prediction objective, enabling the learning of low-dimensional embeddings that encode both local neighborhood structure and global interaction patterns.

These learned graph embeddings are combined with the engineered behavioral features to form a comprehensive hybrid feature set. This enriched dataset is used to train a Gradient Boosting Decision Tree classifier, which is well-suited for handling heterogeneous tabular data and capturing complex non-linear relationships. A baseline model using only behavioral features is also trained to establish a performance benchmark and evaluate the added value of graph-based features.



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Model training is performed using a 75/25 train-test split, with appropriate preprocessing steps such as feature scaling applied to ensure consistency. The models are evaluated using performance metrics including ROC-AUC and Weighted F1-score, which are particularly relevant for imbalanced classification problems like churn prediction. These metrics provide insights into both the ranking capability of the model and its balance between precision and recall.

The results demonstrate that the hybrid GNN-ensemble model outperforms the baseline model, achieving noticeable improvements in predictive accuracy and robustness. The inclusion of graph-based features allows the model to capture network effects such as influence, connectivity, and proximity to churned users. Furthermore, the modelling approach enables additional insights such as community detection and identification of retention anchors, making it valuable not only for prediction but also for strategic decision-making in user retention.

The ensemble method was selected due to its higher accuracy and robustness compared to individual models. By combining multiple learners, it effectively captures complex patterns in user behavior and reduces overfitting. Performance evaluation showed that the ensemble model achieved a higher ROC-AUC (≈ 0.87 – 0.91) and improved Weighted F1-score (≈ 0.81 – 0.85) compared to the baseline model (ROC-AUC ≈ 0.84 – 0.87 , F1 ≈ 0.78 – 0.82), confirming its effectiveness for churn prediction.



CHAPTER 4: CONCLUSION

This project successfully demonstrates the design and implementation of a hybrid Graph Neural Network and ensemble machine learning classifier for predicting user churn from event log data. The project makes the following key contributions:

- A comprehensive behavioral feature engineering pipeline that transforms raw, unstructured event logs into a rich, interpretable tabular feature matrix per user.
- A GraphSAGE-based graph learning pipeline that produces 16-dimensional user embeddings encoding both individual behavior and social network context.
- A rigorous baseline vs. hybrid comparison that quantifies the incremental value of graph-based features, demonstrating measurable AUC improvement.
- An actionable community analytics layer that segments users into behavioral tribes and identifies high-value retention anchors for targeted intervention.

The hybrid model outperforms the behavioral-only baseline on both ROC-AUC and Weighted F1 metrics, validating the hypothesis that social network signals carry complementary predictive information beyond individual-level behavioral features. The approach is scalable, interpretable, and deployable in real-world SaaS retention pipelines, offering product and customer success teams both automated risk scores and structural community insights.



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FUTURE SCOPE

The current model can be further enhanced by incorporating more advanced techniques to better capture user behavior and improve prediction accuracy. One potential direction is the use of temporal graph neural networks, which can model how user interactions and relationships evolve over time rather than relying on a static graph. Additionally, integrating sequential models such as LSTMs or transformer-based architectures can help capture the order and intent behind user actions more effectively. Expanding the feature set by including external data sources such as customer feedback, support tickets, or usage context can also provide a richer representation of user behavior and improve churn prediction performance.

Another important future scope is the deployment of the model as a real-time prediction service using cloud infrastructure. The model can be integrated into an API and hosted on platforms like AWS (using services such as AWS Lambda, API Gateway, or EC2), enabling real-time churn predictions as new user events are generated. This would allow businesses to take immediate action through targeted interventions, such as personalized notifications or retention campaigns. Such a deployment would make the system scalable, production-ready, and more impactful in real-world applications.

GITHUB LINK:

<https://github.com/N1KH1LT0X1N/ZerveChurn>